PROPOSALS

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Contesting for the Post of

General Secretary Technology

Technology Students' Gymkhana

Indian Institute of Technology Kharagpur

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Proposal 1: Alumni Corpus Fund for Permanent Group Projects

Vision

To elevate student-led research under the Technology Committee, Technology Students' Gymkhana, IIT Kharagpur, by overcoming the limitations of strictly operational funding. By establishing an Alumni Corpus Fund, the generated interest income will create a dedicated financial ecosystem to fund the permanent group projects, advanced technology, and capital equipment that are currently out of reach, ensuring long term, sustainable technological innovation.

Objective

The primary objective of this proposal is to engage IIT Kharagpur alumni through structured outreach, encouraging them to contribute to a dedicated Corpus Fund. The interest generated from this fund will be channelled annually towards funding equipment purchases and project development, eliminating dependence on year-to-year budget allocations.

Background & Justification

Research groups under the purview of Technology Committee, Technology Students' Gymkhana face severe financial constraints limiting their ability to procure necessary equipment and develop permanent projects Because the Technology Students' Gymkhana budget is limited and tailored for operational activities, it cannot support these long-term capital needs. A corpus fund model offers a perpetual solution: alumni contribute once, the principal remains invested, and the interest provides a reliable annual income stream. To ensure institutional credibility and smooth disbursement, these funds will be managed and provided through SRIC. IIT Kharagpur alumni have consistently demonstrated their willingness to support technical excellence. With SRIC's professional management and the added incentive of 80G tax exemptions on donations, this model is the most viable path to a sustainable financial ecosystem.

Method of Implementation

- Identify and segment alumni with a strong affiliation to technology societies and research groups through LinkedIn Alumni Outreach and the IIT Kharagpur alumni database.
- Design a structured outreach campaign highlighting the direct impact of their contribution on current students' research and project work.
- Will leverage personal experience and existing, extensive alumni databases from Kshitij to execute a highly focused donor outreach campaign.
- The recent ₹75 Crore commitment by the Class of 2001 validates the high willingness of our alumni to invest in tech initiatives, ensuring this corpus has the potential for massive financial growth.

- The Alumni Office and SRIC will jointly manage the Corpus Fund, directing the generated interest strictly toward funding new technologies and permanent development projects.
- The interest income generated will be distributed among all research groups under the Tech Committee based on a transparent, need-based allocation framework.

Discussion

The Corpus Fund model ensures financial sustainability without burdening the institute's annual budget or requiring repeated fundraising. The principal remains untouched indefinitely, making it a truly perpetual source of funding. Alumni outreach via LinkedIn is a cost-effective and scalable approach, given the large and accomplished alumni base of IIT Kharagpur. The 80G tax benefit further incentivises participation. Over time, as the corpus grows, the annual interest income increases proportionally, benefiting an ever-larger number of research groups and student projects.

Conclusion

The Alumni Corpus Fund is a forward-looking, financially responsible proposal that will transform research group funding from a reactive, year-to-year exercise into a proactive, self-sustaining model. By harnessing the goodwill and resources of IIT Kharagpur's vast alumni network, this initiative will ensure that every research group under the Technology Committee has the financial foundation to pursue excellence in technology and innovation.

Groundwork

- Successfully executed a highly focused donor outreach campaign during my tenure with Kshitij by leveraging an extensive alumni database.
- Initial conversations with stakeholders of research groups have confirmed the critical need for stable, recurring funding for equipment and project development.
- Preliminary meetings with SRIC and the Alumni Office have been completed to establish the operational feasibility of the fund.
- Informal pitches to targeted alumni have been met with enthusiastic support and validation of the initiative's core vision.

Proposal 2: Establishment of a Threadripper Powered High Performance Computing Node

Objective

The primary goal of this initiative is to establish a centralized, high-performance computational backbone for the student technology community at IIT Kharagpur by setting up a dedicated computing node in the Computer and Informatics Centre (CIC). This project aims to provide Inter-IIT contingents and student-led research groups with a 24x7 remotely accessible workstation, eliminating the need for expensive, recurring annual cloud rentals and reducing the dependency on high-latency supercomputing queues.

Background & Justification

Currently, the Technology Committee incurs a recurring annual expenditure of approximately ₹3-4 lakhs on renting cloud computing resources like AWS and Hyperstack, which offers no long-term residual value to the institute. While resources like Param Shakti exist, their significant queue times often hinder the rapid, iterative deadlines required for Inter-IIT Tech Meet Problem Statement. Recent high-stakes projects, such as the Ebullient Securities and LAT Aerospace aerodynamic simulations, have faced critical "time crunches" due to a lack of immediate, high-speed computational power. By investing in a local, professional grade workstation, we can provide a zero latency alternative for student needs, ensuring our teams remain competitive at the highest level.

Method of Implementation

The Technology Committee will procure and maintain a pre-built Lenovo P8 ThinkStation workstation, which is engineered for enterprise-grade reliability and onsite technical support. The system will be housed in a dedicated 6-10 person work unit within the CIC Building, equipped with high speed LAN and a 5-6 Ton HVAC system to manage high thermal loads.

- **System Specifications:** The workstation will feature a 32-core AMD Threadripper 7975WX, 64GB DDR5 RAM, and a specialized NVIDIA RTX Ada 24GB GPU.
- **Remote Management:** To maximize accessibility, the Web Secretaries and Tech Coordinators will implement a remote-access protocol (SSH/VPN), allowing students to submit compute jobs from their hostels and retrieve results without needing physical access to the room.
- **Operational Priority:** During the peak Inter-IIT time, resources will be exclusively reserved for the contingent, while serving as a shared resource for all registered research groups under the Tech Committee throughout the rest of the academic year.

Conclusion

Threadripper is a financially prudent and strategically vital proposal that transforms a recurring rental cost into a permanent institutional asset. With a total investment of ₹11,04,897, the system will achieve a full return on investment within 3-4 years while significantly elevating the quality of research and competitive output at IIT Kharagpur. This project reinforces the institute's commitment to providing students with the professional-grade tools necessary to lead in the domains of robotics, AI, and aerospace innovation.

Groundwork

- I have met with Associate Dean - Innovation & Entrepreneurship and Prof. Prajit Nandi to discuss the feasibility of room allocation in the CIC building and infrastructure requirements.
- Furthermore, I have secured a soft confirmation from Mr. Saket Kumar (Chief of Staff and Founder of Kshitij), who has expressed a strong willingness to facilitate industry connections and alumni outreach to fund the capital expenditure for this workstation.

Proposal 3: Drone Training Centre in Collaboration with 3 Bengal Tech Air Squadron, NCC

Vision

To establish a Drone Training Centre at IIT Kharagpur in collaboration with the 3 Bengal Tech Air Squadron of the NCC, offering IIT Kharagpur students structured drone piloting training and a recognised flying licence, while contributing to India's growing national capability in drone technology and urban aerial operations.

Objective

This proposal seeks to create a formalised drone training programme at IIT Kharagpur, jointly run by the Technology Students' Gymkhana and the 3 Bengal Tech Air Squadron, NCC. The programme will train IIT Kharagpur students in drone piloting and equip them with a recognised flying licence, building a skilled cadre of drone operators within the student community.

Background & Justification

The strategic and operational importance of drones has been decisively demonstrated in recent global events. Operation Sindoor highlighted the critical role of drone technology in modern Indian defence operations, while the ongoing conflict in Ukraine has established unmanned aerial systems as indispensable tools in urban warfare and surveillance. India is rapidly expanding its drone ecosystem, with significant policy support, manufacturing incentives, and defence procurement focused on indigenous drone development. IIT Kharagpur, as a leading technology institute with an active NCC unit and established research groups, is uniquely positioned to be a pioneer in student drone training. Establishing this centre would not only develop individual student skills but also contribute meaningfully to India's national drone capability.

Method of Implementation

- Formalise a Memorandum of Understanding (MoU) between the Technology Students' Gymkhana, IIT Kharagpur and the 3 Bengal Tech Air Squadron, NCC for joint conduct of the Drone Training Programme.
- Design a structured training curriculum covering drone piloting fundamentals, flight regulations, safety protocols, and practical flying sessions, culminating in a recognised flying licence.
- Leverage the NCC's existing infrastructure, airspace permissions, and regulatory relationships to ensure compliance with DGCA (Directorate General of Civil Aviation) norms for drone operations.

- Conduct training in batches for IIT Kharagpur students and recognised DGCA-compliant flying licence will be provided to students who successfully complete the training programme.

Discussion

The collaboration with the 3 Bengal Tech Air Squadron, NCC provides institutional legitimacy, access to trained instructors, regulatory experience, and airspace approvals that would be extremely difficult to obtain independently. IIT Kharagpur's existing strengths in aerial robotics research provide a strong academic foundation on which practical piloting skills can be built. A certified drone pilot programme at an IIT would be a nationally significant initiative, attracting attention from defence establishments, drone manufacturers, and government bodies. In the long run, trained students could contribute to indigenous drone development, defence R&D, and the growing commercial drone services sector in India.

Conclusion

The Drone Training Centre is a forward looking, nationally relevant initiative that positions IIT Kharagpur at the forefront of India's drone revolution. By partnering with the 3 Bengal Tech Air Squadron, NCC, the institute can offer students a unique, certified skill that has both defence and civilian applications. This proposal aligns with India's national priorities in drone technology and reflects IIT Kharagpur's commitment to producing graduates who can contribute meaningfully to the nation.

Groundwork

- The 3 Bengal Tech Air Squadron, NCC at IIT Kharagpur have been approached for collaboration on the programme structure.
- DGCA drone pilot certification requirements have been reviewed to ensure the proposed curriculum leads to a recognised licence.
- Strong student interest in a Drone Training Centre has been validated through direct conversations with Mr. Mithun Chowdhary, confirming significant demand within the student community and a Kolkata-based drone company, has been identified as a potential partner for funding support and provision of drone hardware and facilities for the programme.

Proposal 4: Paperless Digital Workflow System

Vision

To eliminate paper-based bureaucracy within the Technology Students' Gymkhana (TSG) and its interface with institute offices and administrative sections by implementing a comprehensive, paperless digital workflow system. This platform will be developed and supported by the Technology Students' Gymkhana's Technology Coordinators and Web Secretaries, thereby establishing administrative processes that are transparent, trackable, and efficient.

Objective

To replace the time-consuming physical file system within the Technology Students' Gymkhana by deploying an in-house digital platform. This initiative aims to digitize the end-to-end lifecycle of paperwork, ensuring a paperless, transparent, and accelerated approval process among the Technology Students' Gymkhana, student bodies, and the Institute Administration.

Key Features & Functionality

- **Centralized Event Approvals:** A streamlined digital workflow for submitting and processing event permissions with institute offices.
- **Integrated Financial Processing:** A dedicated module for efficient handling of society funding, budget approvals, and reimbursements.
- **Real-Time Status Tracking:** A dashboard allowing students of IIT Kharagpur to monitor the live progress of their files without physical tracking.

Background & Justification

The current paper-based administrative processes within the Technology Students' Gymkhana and between the Technology Students' Gymkhana and institute offices result in significant delays, a lack of transparency, difficulty in tracking file status, and an unnecessary environmental impact. Files passing through multiple desks lose time and accountability at every step. A digital platform would resolve all of these issues simultaneously: approvals become faster, processes become transparent, files can be tracked in real-time, and the institute moves closer to a paperless, environmentally responsible model. Given that the Dean of Administration's approval for this initiative is currently in process, the institutional backing for this proposal is strong.

Method of Implementation

- **Team Formation:** The management of the initiative will be handled by the General Secretary Technology, along with the Technology Coordinators and Web Secretaries. If additional support is required beyond the student executive council for the actual

development of the platform, a dedicated technical team comprising skilled students of IIT Kharagpur will be formed.

- **Platform Development:** Design a comprehensive digital platform that digitizes the following workflows: event permissions and approvals, society/club funding requests, GC and Open IIT registrations, Inter-IIT Tech Meet documentation, budget approvals, and reimbursements.
- **Tracking Dashboard:** Implement a real-time file tracking dashboard so that students of IIT Kharagpur, societies, and administrators can monitor the status of any submission at any given time.
- **Phased Rollout:** Deploy the system in a phased manner: first within the Technology Students' Gymkhana internally, then gradually extending to inter-office workflows with the institute administration.
- **Incentives:** Students of IIT Kharagpur and faculty members who contribute significantly to the development and deployment of this system will receive a Letter of Recommendation from the General Secretary Technology to acknowledge their high-quality participation.

Discussion

Building this digital platform in-house leverages the considerable technical talent within IIT Kharagpur and avoids dependence on external vendors. Security, accessibility, and overall accountability for the platform will rest with the Technology Students' Gymkhana, managed directly by the concerned council members. The LOR incentive ensures that the best technical minds are motivated to contribute. The phased deployment approach minimizes disruption, allowing the system to be tested and refined before a full-scale rollout. Over time, this system can serve as a model for other Technology Students' Gymkhana bodies and the broader institute administration.

Conclusion

The Paperless Digital Workflow System is a long-lasting administrative reform that will benefit every student, society, and administrative section interacting with the Technology Students' Gymkhana. It will save time, increase transparency, reduce environmental impact, and modernize the administrative infrastructure of IIT Kharagpur. With the Dean of Administration's approval in process and strong in-house technical capabilities, this proposal is both timely and highly feasible.

Groundwork

- **Administrative Consultation:** Extensive discussions have been held with the Dean (Administration) and various administrative heads. These interactions highlighted a critical pain point: the current reliance on manual file tracking leads to delays and

bottlenecks in inter-office movement. There is a consensus on the need for a digital transition to streamline these workflows.

- **Stakeholder Analysis across Sections:** Consultations were conducted with the Technology Students' Gymkhana Office, Society Office-bearers, and Club Heads. These sessions identified that the primary hurdle for student bodies is the repetitive nature of paperwork for room bookings and event permissions. The feasibility of a "One-Time Submission" model for recurring files was validated through these discussions.
- **Technical Validation:** Preliminary meetings with the Technology Coordinators and Web Secretaries have confirmed the technical viability of building an in-house platform.
- **Student Outreach & Feedback:** Direct interactions with the general student body revealed a strong demand for a transparent, real-time tracking system.